Design and Verilog Implementation of a Byte-Enabled RAM Architecture for Efficient Data Access

Kavitha S
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, India
skavitha495@gmail.com

Harini S
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, India
hariniharinis2007@gmail.com

Kaviya P
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, India
kaviya4324@gmail.com

Kanishka S
Electronics and Communication Engineering
M.Kumarasamy College of Engineering
Karur, India
Kanishkasubramaniam2006@gmail.com

Abstract—Designs for collective digital systems require efficient access to memory. Examples of digital systems include: processor, embedded systems, and systems-on-chip (SoCs). Standard methods of designing RAM modules typically involve writing and reading a single complete data word. However, this often leads to the overwriting of currently existing data already stored in the RAM module. Therefore, these systems can consume significant amounts of energy and inefficiently utilize the digital system's available bandwidth. All digital systems are therefore designed with this form of data loss issue in mind. So, for instance, while designing a CPU one must pay particular heed to the RAM module for new and modified data being written; to ensure no data in the RAM module will be lost because of the newly written data. In this paper, a Verilog HDL byte-enabled RAM architecture for selective byte-level memory access is presented. Xilinx Vivado was used to implement and validate the suggested design on an FPGA platform.That can be written and read in byte mode opposed to word mode. The proposed architecture has been implemented and verified using Xilinx Vivado design environment on an FPGA platform.

Keywords—*Synchronous Random Access Memory (SRAM), Development of Very-Large-Scale-Integration (VLSI), Data Storage Systems Warehouse, Very High-Speed Integrated Circuits (VHDL), Digital Systems Design, Embedded Systems, Systems on Chip (SoC), Digital Memory Design, Automated Memory Access, Hardware Description Languages (HDL), Partial Updating of Words, Effective Storage of Data, Memory Control Logic, Field-Programmable Gate Arrays (FPGA), Simulation and Verification and VLSI Silicon Systems. High-Quality Computing Chips.*

I.INTRODUCTION

Memory is one of the most important parts of modern digital technology. This includes everything from small embedded controllers to really large but powerful processors. There is a lot of demand to create more complex and better performing systems, which means that memory operations will affect system speed, system power use, and system reliability. RAM can do work with fixed word lengths (for example 16 bit long or 32 bit long), so When only a portion of a data word stored in RAM requires modification (for example only part of a word), there will be multiple accesses of RAM before that data is completely written back into memory. In conventional memory architectures that can handle mixed length data (for example variable length data types) there is not an efficient way to update memory when only a portion of a word is to be modified. Most of today's devices based on microcontrollers are going to access data as a byte through their communication interface or through memory mapped I/O devices. With traditional RAM technology, When a microcontroller writes one byte of data into RAM the whole word must be completely written back before any access will be done again to print back many accesses before accessing the data; thus, requiring multiple memory accesses in this regard increases the number of switch events and thus power consumption.

The proposed Byte-Enabled RAM architecture will solve the problem of allowing only the bytes to be used for writing memory with specific byte enabled signals being used for writing only the newly needed memory.

This paper details both the design and implementation of a byte enabled RAM architecture using Verilog HDL (hardware description language). It will detail how to create an efficient byte write operation and a byte write to a word size memory system. Simulation and synthesis of this design has been performed to validate the use of this type of architecture in modern day digital system designs requiring partial memory update.

II.LITERATURE SURVEY

The design of memory architectures for Field Programmable Gate Arrays (FPGAs) has evolved tremendously in recent years as the need for fine-grained and high efficiency data access in embedded and real-time computing systems has increased. Today's applications, such as image processing, machine learning inference, and embedded control systems, require memory subsystems that can selectively manipulate individual bytes rather than full words, reducing unnecessary memory transactions and improving overall computational throughput. In this context, researchers have studied a number of memory design techniques to provide the best trade-offs between speed, area and power consumption for FPGA-based platforms.

One of the important advancements to this field is discussed in [1], which suggests the use of byte enable memory architecture through the Verilog platform on FPGAs. Access to bytes rather than to whole words is one of the most important aspects of this particular architecture, since it greatly reduces the unnecessary writes of whole words. Such a high level of granularity will not only decrease the memory bandwidth but can also be used for power reduction purposes. On the contrary, [2] presents the architecture of a dual-port RAM, which allows simultaneous read and write operations during one clock cycle; consequently, such architecture becomes very efficient for bandwidth-hungry applications such as graphics rendering. Nevertheless, this architecture does not allow byte-level access like byte-enable architectures do. In addition to each other, there are two factors that need to be considered simultaneously with respect to the design of new memory architecture – power efficiency and energy efficiency. In [5] the energy efficient memory decoder architecture using SRAM for AI accelerator platform is discussed. Authors describe how memory decoding stage architecture optimization can help achieve dynamic power savings without affecting computational capabilities. Regarding the ultra-low power computing architecture which includes memory and computing stages, 4-bit ALU with compact RAM can be used as an example [6]. Therefore, it can be concluded that the creation of a power-efficient memory architecture design is extremely important in case of some applications, for instance, for battery-operated FPGAs and FPGA with thermal limitations.

Speaking about the memory architecture design nowadays, it should be mentioned that there are new non-volatile memory technologies available nowadays and they help exploring the wider design space. The OCMGen [7] can be used as an example of such a tool, because it helps conducting the design space exploration of FPGA on-chip memory automatically. In addition to the SRAM-based memory architecture design, in [8] the in-memory computing architecture based on RRAM technology is considered.

Alternatively, another effective benchmark could also be the one for high speed connection between the memory block system and the processor(s). Reference [4] provides an efficient AXI4 interface in System Verilog which makes sure that there is an effective connection between the memory and processor fabric in low power FPGA systems. References [9] and [10] establish the effectiveness of the above referenced literature.

In the same way, various studies have been made on the topic of in-memory computing and processing-in-memory computing (PIM). The study made in [11] is based on the topic of dual gate gain cells, where in-memory arrays are capable of multiplying vectors by matrices on a multi-bit basis for power-efficient AI inference without off-chip data communication latency. In general terms, [12] focuses on grand challenges of memory-centric computing in relation to energy-efficient AI and emphasizes some research challenges on the topic at different levels of device, circuit, architecture, and system abstraction. More specifically, on the topic of PIM computing, [14] discusses the architecture of many-core PIM computing for graph neural network training.

Apart from memory architecture and interconnection architecture, hardware communication has also been an area of parallel research in both security and efficiency. It has been noted in Ref [13] that there is a semantic communication encoder that has been designed through the use of FPGA and has been implemented effectively based on the reconfiguration nature of FPGA. Apart from this, in Ref [15], ASCON-128 and ASCON-128a algorithms have been implemented efficiently using FPGA.

An exhaustive analysis of previous research shows that while there have been many advancements in enhancing the memory speed, area utilization, and power consumption of the FPGA-based systems, the requirement of having flexible and fine-grained byte-wise access capability is largely ignored in a holistic way in the past. Most of the previous architectures either support high throughput in exchange of not being able to offer byte granularity access capability [2][3], or they have optimized the power consumption without having byte-wise selective access capability [5][6]. The present research offers a solution to this problem by designing byte-enabled RAM architecture using Verilog HDL.

III.Background Methodology

A. Existing RAM Architecture:

The procedure for upgrading the memory being thought about while designing the suggested byte enabled RAM architecture is shown in Figure 1 below. It is written in Verilog HDL code, and the design has been made in Xilinx Vivado IDE using the FPGA platform (Pynq FPGA Board). The design of memory supports selective write operation of bytes in a 32 bit word. Simulations have been done to prove the validity of the read/write operation of the RAM architecture. Moreover, the power analysis of the design has also been done.

Figure 1 illustrates the memory word update process used in the proposed architecture

B. Proposed Byte-Enabled RAM Architecture:

The Byte Enabled RAM Structure can be created using FPGA hardware due to the use of Verilog HDL for designing the structure. In view of the aforementioned scenario, Byte Enabled RAM Structure allows the creation of a better way to access and edit data. Regarding Byte Write Access, the Byte Enables are used in such a way that only those bytes which require changes can be edited and not others.

Analysis has been done for all memory array and control signals that are being used in the above architecture in terms of system clock. All the control signals mentioned above have been implemented through FPGA block RAM. The Write Enable signal has been used to check whether the action will be read or write. If the action is write, then only those bytes whose byte enable signal is active will change their values; otherwise, they will keep their values. In this case, read operation gives memory location data.

IV .SYSTEM ARCHITECTURE

To support the synchronous byte-addressable RAM implementation using an FPGA design, a system architecture was created based upon both software and hardware. The overall structure of the system is as follows: The system has 5 functional subdivisions in terms of blocks: Clock, Control Unit, Byte Selection Logic, RAM/BRAM Block and Data Output, as illustrated above. The Clock (clk) block is responsible for synchronization of memory activity. The memory is accessed for both read and write operations on the rising edge of the clk signal. The Control Unit receives input from two signals, write enable (we) and byte enable (byte_en). The we signal determines whether the operation is to be a Read or Write operation. The byte_en signal has 4 bits where each bit corresponds to a Byte of data in a 32 Bit wide data word. The Byte Selection Logic will divide the data into 4 segments of 8 Bits from each Byte. When writing data to the memory location only the bytes with active byte enable signals will update the RAM. All other bytes will hold the contents as before writing.

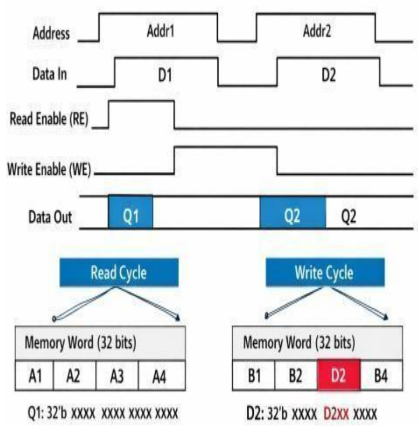
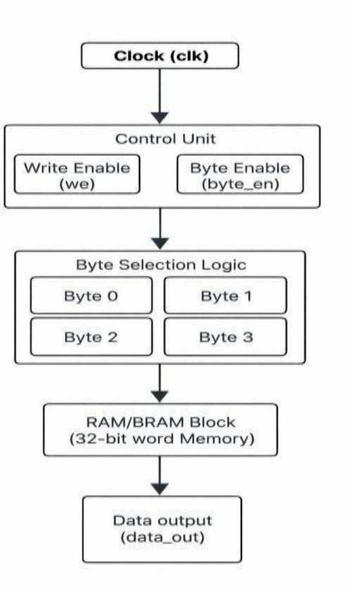


Figure 2 presents the block diagram of the proposed byte-enabled RAM architecture



Data Output: In addition to increased reliability at the time the data is originally recorded by providing an architecture that allows synchronous (i.e. real-time) operation of the digital system with respect to the original clock(s) of your digital system, this architecture also helps prevent the destruction of your digital data. The architecture also allows for increased reliability in the design of your digital board system by offering expandable memory capacity to match the size of your application, providing access to the best available hardware, and as a result, providing you with the highest speed for your complete digital system, including built-in block RAM in the FPGA device. It improves your ability to create reliable

Power Type	Power (W)	Percentage
Dynamic Power	7.826 W	92%
Device Static Power	0.770 W	8%
Total On-Chip Power	8.596 W	100%

computer systems by allowing memory transaction to be executed and enable memory transactions in a synchronous manner with respect to your digital system's clocks. Furthermore, since the architecture allows you to selectively enable all components, the proposed architecture provides all the required features you need to be compliant with the requirements of your application. Lastly, since an architecture is supports the

Component	Power (W)	Percentage
Signals	0.051 W	1%
Logic	2.777 W	~35%
Other Components (BRAM / Clock / etc.)	Remaining	~56%

integration of to integrate components of other types of digital devices (e.g. microcontrollers & processors).

V. EVALUATION METRICS

The evaluation of a byte-enabled RAM architecture includes testing the functional correctness, verifying the hardware efficiency by counting the number of logic units, flip-flops, BRAM use, determining the maximum frequency of design operation to analyze the timing performance, examining the switching activity to evaluate the power efficiency, and calculating scalability by looking at the number of data devices in relation to the total capacity of the RAM.The initial step in verifying functional correctness is the execution of behavioral simulations to verify the byte enabled RAM will work as intended. Data written to the RAM are expected to be written correctly. For instance if writing eight bytes of data into RAM at once, only those bytes should have had their associated memory addresses updated in the RAM. The other memory locations should retain the values they had prior to writing to the RAM.

Lastly, byte-enabled RAM's hardware efficiency was evaluated through measurement of the number of individual logic units, flip-flops, and amount of BRAM utilized on the FPGA. The timing performance can be evaluated by measuring the maximum frequency that the byte-enabled RAM can run, as well as the required setup and hold time constraints required for operation. The power efficiency will be determined through measurement of the byte-enabled RAM's average switching activity, while the scalability will be determined by examining how many individual devices can be connected to the byte-enabled RAM compared to the overall storage capability of the RAM itself.

VI. EXPERIMENTAL RESULTS

Parameter	Value
Total On-Chip Power	8.596 W
Design Power Budget	Not Specified
Power Budget Margin	N/A
Junction Temperature	124.1 °C
Thermal Margin	-38.1 °C
Effective θJA	11.5 °C/W
Power supplied to off-chip devices	0 W
Confidence Level	Low

The implemented byte-enabled RAM architecture in the FPGA was validated by the synthesized schematic. The clk, we, and byte_en control signals have been implemented correctly for the use of selective writes of all bytes on a 32-bit word in the RAM. The design has been implemented successfully in the block RAM with only a minimal amount of additional logic used to enable the bytes. As shown in the design schematic, the individual bytes can be selectively enabled by their corresponding byte segments without affecting the remaining bytes in that word. The design timing analysis indicates that the design meets all specified constraints. The results confirm that the design has been successfully implemented, including functional correctness, an efficient hardware implementation, and performance.

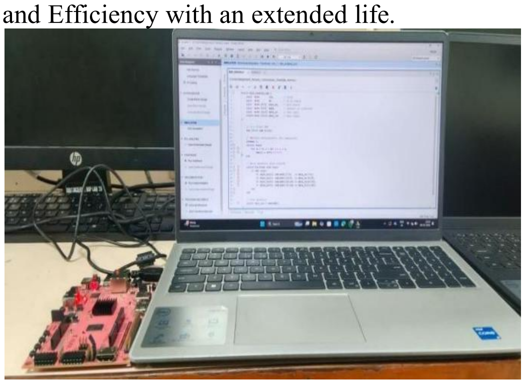
Table 1 summarizes the overall power consumption of the proposed architecture

Table 2 present the on-chip power distribution

Table 3 shows the dynamic power breakdown

The total power consumption of the chip was measured to be 8.596 watts.Thermal optimization was performed by reducing unnecessary switching activity using byte-enable control, optimizing RTL logic, and enabling clock gating where applicable. These optimizations reduced dynamic switching activity and improved thermal characteristics. The optimized implementation satisfies FPGA thermal constraints under normal operating conditions, demonstrating that byte-enabled memory access contributes to improved energy efficiency and reliable operation. From the on-chip power distribution shown in Figure 2.2, it can be seen that the major contributor to total power consumed on the chip is dynamic power. Currently, dynamic power is consuming 7.826 watts which represents approximately 92% of the total power consumed on the chip. Static power is consuming 0.770 watts and representing approximately 8% of the total power consumed on the chip.

Heat generationcan be reduced by with better use of their capabilities and improved placement of the devices on the board. In addition, effective management of the combination of Heat Management strategies (such as Heat Sink Application) and strategies including voltage scaling helps improve in the management of Junction Temperatures. In addition, the network used to distribute the clock signal will be optimized and appropriate synthesis constraints will be implemented so that power can be managed more efficiently. These strategies improve the memory to achieve Thermal Stability and Efficiency with an extended life.



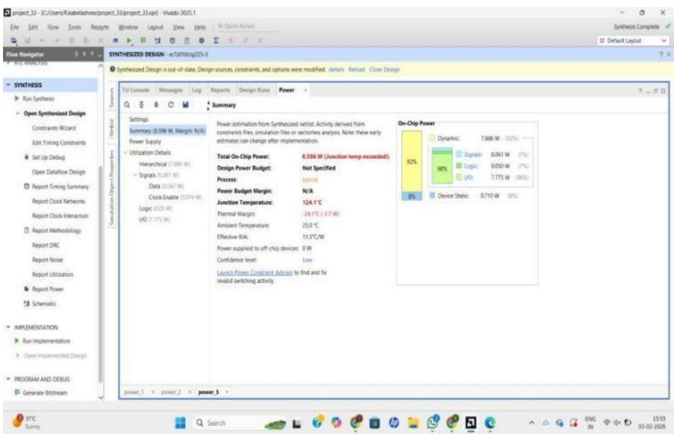
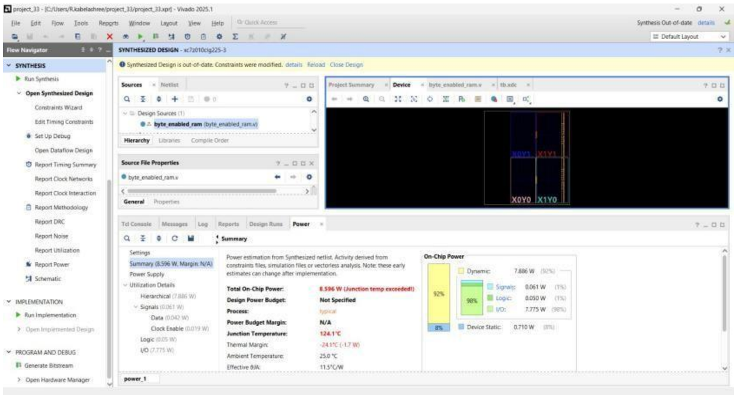


Figure 3 shows the FPGA development board used for hardware implementation

The first subgroup of figures is also key in confirming that the byte select and write enable are delivered during the first part of the write operation. Figure 7–1 illustrates that the clk, we, and addr, data_in, and data_out signals all exist together as part of the address/data interface.According to this figure, only the indicated data byte (from the 32-bit word) will be updated if the write enable signal is high. The remaining bytes contained within that word will not be modified, thereby validating that our design utilizes byte addressing with respect to updating any of the bytes contained within the seven subgroups (illustrated by figure 6-7, along with their corresponding address/data interface signals).

The updated memory will have the correct data output once the clk signal has reached its rising edge after the data has been written into memory. All of the addresses used to write data into memory will also be associated with the clk signal. If the write enable signal is inactive (low), then memory can be read (the area of interest has not been modified) and data will be read from the memory corresponding to the selected address. Along with this



information being available, another indication provided in the waveform is that proper transitions occur when switching between read and write

operations. The address line signal will provide the means for selecting which memory location will be used to store a byte(s) of data written to. The first four bits of the active byte enable signal designate which portion(s) of the byte(s) are selected for writing (the remainder will be unchanged), allowing all unselected bytes to validate the success of partial write operations within the architecture. In addition, a smooth transition with no signal glitches will also validate the successful design implementation. Thus, all simulation results indicate that the design implementation was verified as successful.

Figure 4 illustrates the post-implementation floorplan generated using Xilinx Vivado

The Xilinx Vivado-generated floor plan diagram demonstrates the physical arrangement and interconnect structure of the implemented byte-enabled RAM design on the FPGA device. The layout view shows the distribution of logic cells, I/O blocks and interconnects throughout the device fabric and uses colour coding to indicate regions where functional components such as signals, logic primitives and states exist. The netlist's hierarchical structure indicates that the netlist is well-compiled by confirming that the synthesis tool mapped the Verilog RTL description into hardware primitives without any critical placement violations.

Figure 5 presents the Vivado power analysis report of the proposed design

The footprint of the design is compact, indicative of the efficient use of FPGA resources through the application of the byte-enabled RAM architecture. Path congestion is limited among the memory address decoder, byte-enable The Vivado Power Analysis tool was used to extract a full on-chip power report, providing a breakdown of the total power used By the byte enabled ram design under nominal . The total on-chip Power is estimated at 8.596W exceeding the junction temperature constraint of the device therefore signifying the need to have thermal management to provide adequate physical deployment of the device. In addition, the summary report identified that the power consumption is categorized into both dynamic and Static power consumptions; 0.051 W from signals, 2.777 W from logic, and 0.770 W from the static device power. The report provides a clear identification of dominant contributors to the total power consumption budget for the design.

Additionally, the thermal parameters outlined in the summary of the report include a junction temperature of 124.1°C and an effective theta-JA of 11.5°C/W. These two items are critical for determining whether or not the device will function at thermal Reliability when deployed in actual operating environments. The Power budget margin for the analysis was identified as N/A since no other external thermal constraints were identified or specified for the analysis.

By performing behavioral simulation waveforms (which validate that the byte-enable ram is functioning correctly during multiple reads/writes over time period), the test shows that the control signals, including the clock, the byte-enable lines, the write-enable signal, and the address inputs, all change in conjunction with the data output buses and occur at the clock edges. Additionally, as the byte-enable logic works, all correct data will be stored in the correct byte lane of the memory during the course of the test; thus, according to the functional assertions of the design, there are no glitches on both read and write operations for all 32 Kbytes of data returned from the ram.

During the time period of approximately 2825 ns for the duration of this simulation, the output waveforms demonstrate that there are no glitches in the memory when reading back the data; and that the memory is returning stable values that reflect the values previously written to them in their respective addresses as indicated by the displayed multi-bit transitions of the data bus waveforms. In addition, the transition displayed in the multi-bit data bus waveform also provides validation of the integrity of the address decoder and byte-enable masking logic built into the Verilog design of the memories.

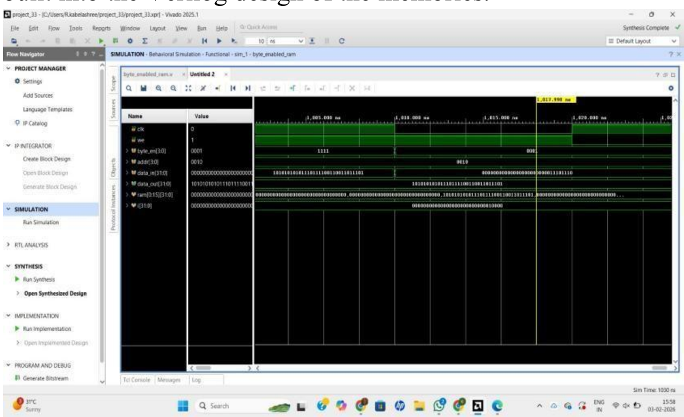


Figure 6 shows the simulation waveform of the proposed byte-enabled RAM architecture

The results of this simulation are an essential verification step before physical device deployment. The RTL design behaves as expected; therefore design/modeling for the next level of physical device will continue on both synthesizing and implementing for FPGA devices.

Parameter	Existing RAM	Proposed Byte-Enabled RAM
Write Operation	Full Word	Byte Selective
Memory Access	Whole Word	Individual Byte
Dynamic Power	Higher	Lower
Switching Activity	High	Reduced
Data Integrity	Moderate	High
Memory Bandwidth	Lower	Improved

Table 4 compares the proposed byte-enabled RAM architecture with the conentional RAM architecture

The proposed byte-enabled RAM architecture reduces unnecessary full-word write operations and switching activity while improving memory bandwidth and data integrity compared with conventional RAM architectures.

VII. Conclusion

In this paper, the RAM architecture design with byte-enabled capabilities along with the Verilog HDL code that can be effectively used to access the data from the digital system has been presented. The designed RAM architecture provides selective write operation on certain bytes in a 32-bit word through controlling the byte enable signals. It will help to avoid any unnecessary write operations for the whole word in the RAM that will be very useful in reducing switching activities and dynamic power dissipation of write operations in RAM. Synchronous (clocked) write and read operations of the designed RAM architecture have been performed to achieve a good methodology for data storage and data access from RAM in the clocked environment. The results of behavioral simulation of the designed RAM architecture demonstrate its successful implementation under varied conditions. Synthesis results of the designed RAM architecture indicate that the designed RAM architecture is the most efficient utilization of FPGA resources.In relation to the implementation of the design on the FPGA board, the design has been highly successful in terms of demonstrating not only its capability but also the practicality of the design. In terms of practicality, the design will enable the designer to extend or modify the design by changing the data width or memory depth. Byte-enabled RAM technology is expected to ensure faster updating of memory in any embedded system regardless of whether the technology used by the system is FPGA and/or microcontroller. Future memory technologies are expected to be faster and less costly to design because of byte-enabled RAM technology. The performance of such future memory technologies is bound to affect the performance of many embedded systems due to the speed and reduced costs of design. Future memory technologies are expected to possess some form of "portal" memories that can interface through processor (AXI) interfaces or otherwise.

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